

This phase involves conducting a detailed study of existing agricultural recommendation systems, crop prediction technologies, and machine learning-based smart farming solutions. Various research papers, journals, case studies, online resources, and agricultural technology platforms are analyzed to understand how artificial intelligence and data analytics are currently being used in agriculture.

The literature survey focuses on understanding different machine learning algorithms used for crop recommendation, yield prediction, soil analysis, and environmental monitoring. Existing techniques such as Decision Trees, Random Forest, KNN, Logistic Regression, Neural Networks, and clustering algorithms are studied to compare their performance, advantages, and limitations in agricultural applications.

The survey also examines data preprocessing techniques, feature engineering methods, dataset balancing approaches, and evaluation metrics commonly used in agricultural machine learning systems. Existing smart farming platforms and precision agriculture technologies are analyzed to identify gaps, challenges, and opportunities for improvement.

This phase helps in understanding current technological advancements, identifying research limitations, and selecting suitable methodologies for developing an efficient agricultural production optimization system. The insights gathered from the literature survey form the foundation for model development, system architecture, prediction optimization, and future agricultural innovations.